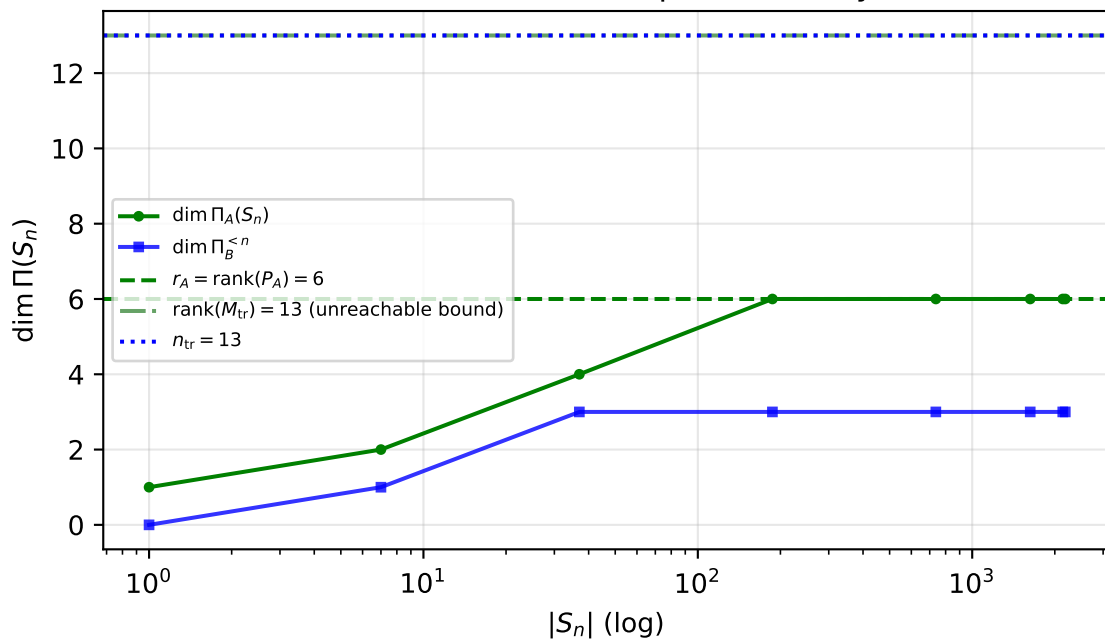
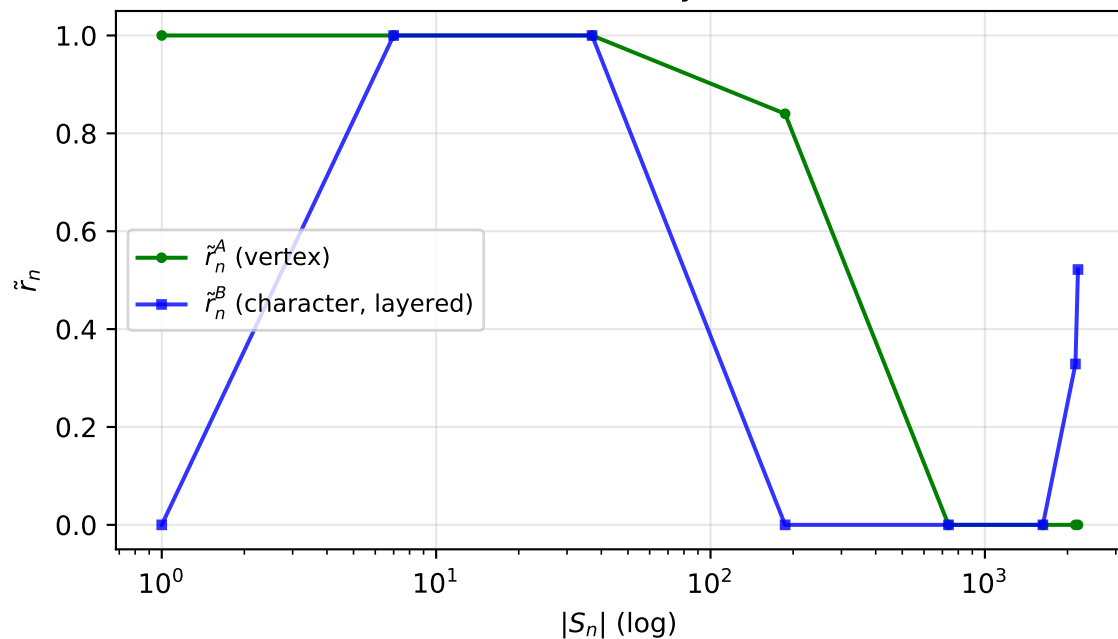


Version A vs B (character-based) on $X^{5,13}$, shell-layered cascade
PGL(2,13): $|G| = 2184$, $n_{\text{tr}} = 13$, $\text{rank}(M_{\text{tr}}) = 13$, $r_A = \text{rank}(P_A) = 6$

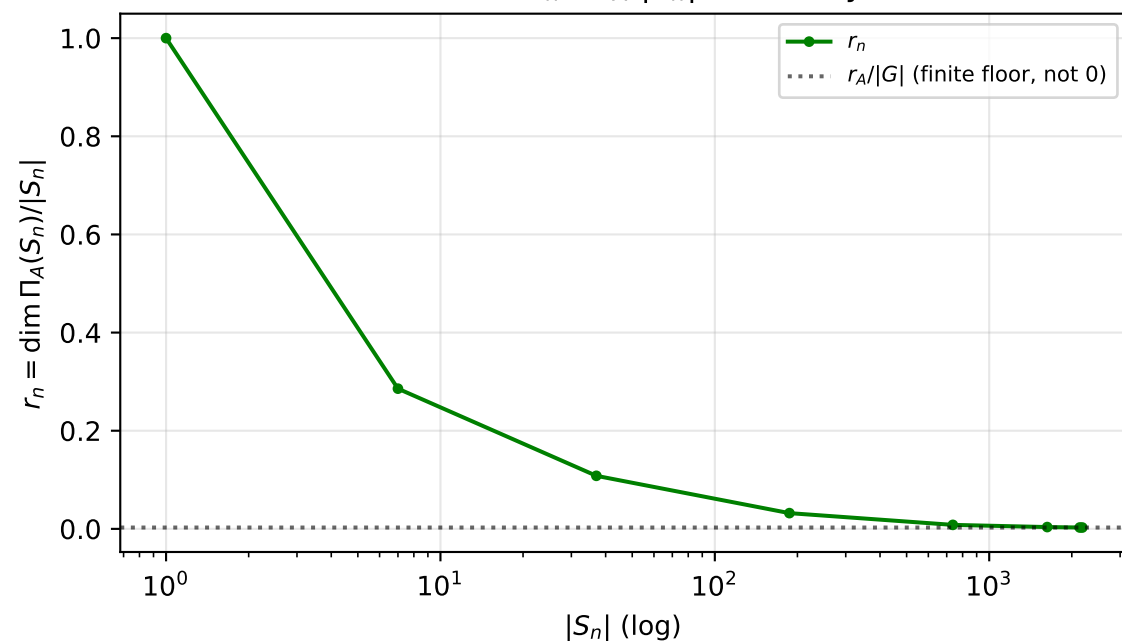
Rank of character-trace span (shell-layered)



Shell-wise novelty fraction



Finite bound $r_n \leq r_A/|S_n|$ (Corollary~1)



Summary

KEY NUMBERS ($q=13$, $p=5$), shell-layered:

Group: PGL(2,13)
 $|G| = 2184$
Generators in 1 conjugacy class(es)
 $n_{\text{tr}} = 13$ of 15 classes
 $\text{rank}(M_{\text{tr}}) = 13$
 $r_A = \text{rank}(P_A) = 6$ (7 trace-selected sectors have $\kappa_p = 0$)

VERSION A (empirical, this traversal only):
 $\dim \Pi_A(S_n)$ reaches r_A at $|S_n| \approx 187$

VERSION B (character, layered):
Max $\dim \Pi_B^{\leq n}$: 3
Late $\tilde{r}^B$: 0.170